MOOC_3.6_路由表分析实验

路由表分析实验

- 实验目的
- 实验内容

路由表

[Quidway]display ip routing-table

Routing Tables:

Destination/Mask	proto	pref	Metric	Nexthop	Interface
0.0.0.0/0	Static	60	0	120.0.0.2	Serial0
8.0.0.0/8	RIP	100	3	120.0.0.2	Serial0
9.0.0.0/8	OSPF	10	50	20.0.0.2	Ethernet0
9.1.0.0/16	RIP	100	4	120.0.0.2	Serial0
11.0.0.0/8	Static	60	0	120.0.0.2	Serial0
20.0.0.0/8	Direct	0	0	20.0.0.1	Ethernet0
20.0.0.1/32	Direct	0	0	127.0.0.1	LoopBack0
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20.0.0.1/32	Direct	0	0	127.0.0.1	LoopBack0

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静态路由

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0.0.0.0/0	Static	60	0	120.0.0.2	Serial0
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9.0.0.0/8	OSPF	10	50	20.0.0.2	Ethernet0
9.1.0.0/16	RIP	100	4	120.0.0.2	Serial0
11.0.0.0/8	Static	60	0	120.0.0.2	Serial0
20.0.0.0/8	Direct	0	0	20.0.0.1	Ethernet0
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动态路由

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20.0.0.1/32	Direct	0	0	127.0.0.1	LoopBack0

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直连路由

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9.0.0.0/8	OSPF	10	50	20.0.0.2	Ethernet0
9.1.0.0/16	RIP	100	4	120.0.0.2	Serial0
11.0.0.0/8	Static	60	0	120.0.0.2	Serial0
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20.0.0.1/32	Direct	0	0	127.0.0.1	LoopBack0

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20.0.0.1/32	Direct	0	0	127.0.0.1	LoopBack0

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路由表示例

数据包的目的IP地址：**9.1.1.2**

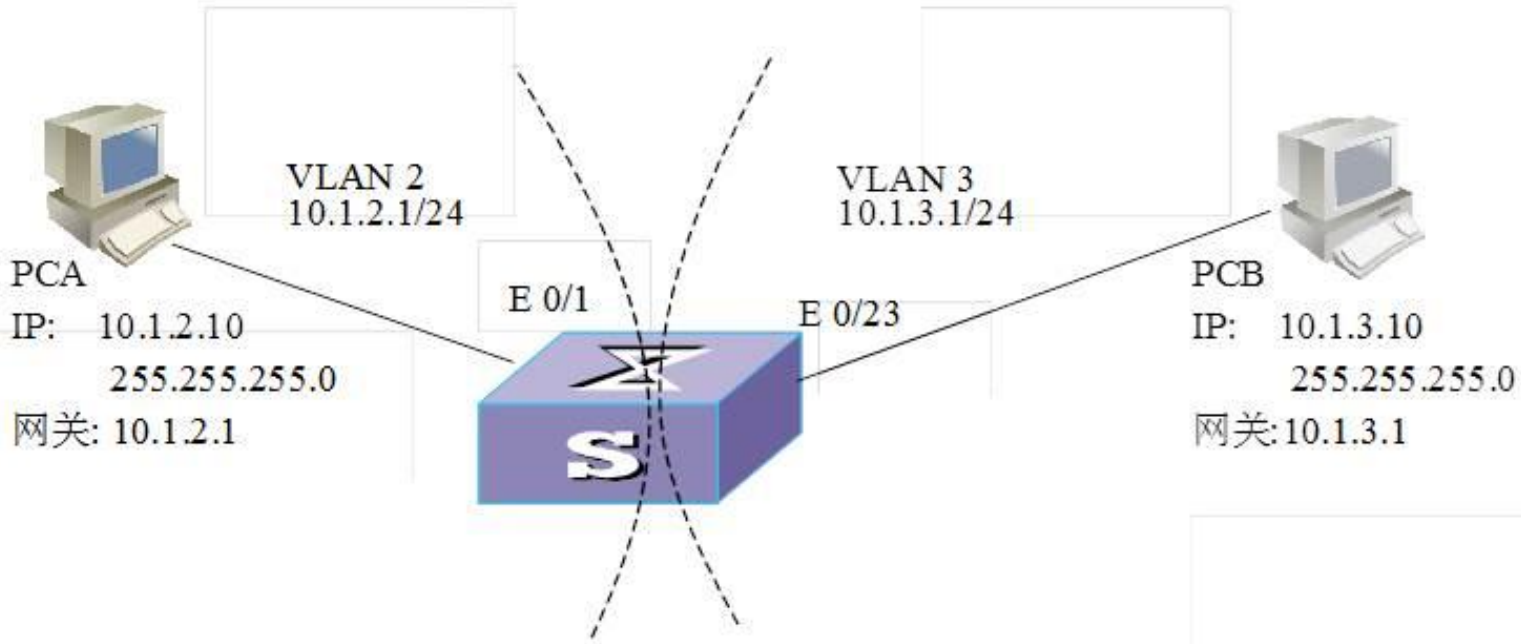
[Router]display ip routing-table

Destination/Mask	proto	pref	Metric	Nexthop	Interface
0.0.0.0/0	Static	60	0	120.0.0.2	Serial0
8.0.0.0/8	RIP	100	3	120.0.0.2	Serial0
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20.0.0.1/32	Direct	0	0	127.0.0.1	LoopBack0

- 路由表查询：　最长匹配原则

目的IP和子网掩码作“与”运算获得网络地址

组网图


VLAN 2
10.1.2.1/24
VLAN 3
10.1.3.1/24
PCA
IP: 10.1.2.10
255.255.255.0
网关: 10.1.2.1
E 0/1
E 0/23
PCB
IP: 10.1.3.10
255.255.255.0
网关:10.1.3.1